ftp

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```r
library(dplyr)
library(tidyverse)
library(ggplot2)
df <- read_csv("LendingClub_Data.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

Clean/Prepare Data for Analysis

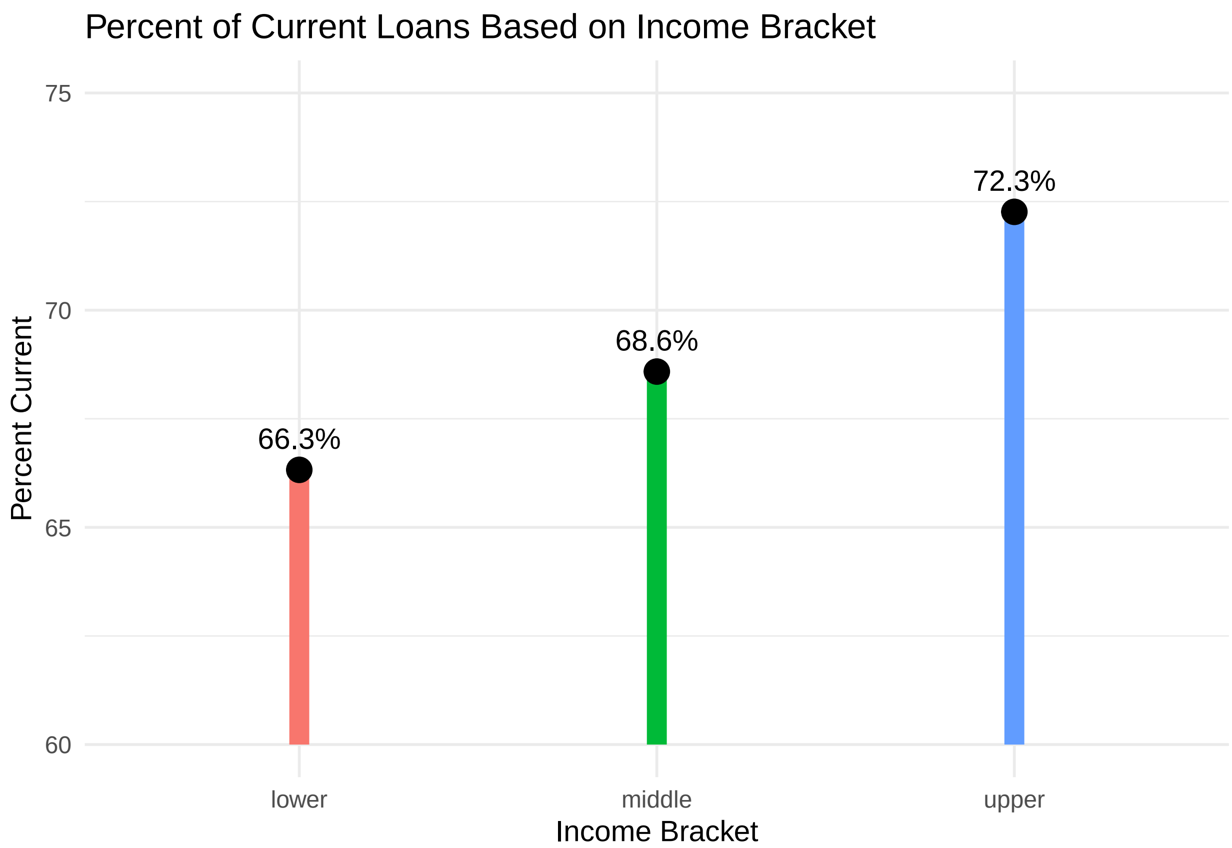
```r
# Filter and Clean
df_clean <- df |>
  select(home_ownership, loan_status, annual_inc, emp_length) |>
  filter(!is.na(home_ownership), !is.na(loan_status), !is.na(annual_inc), !is.na(emp_length)) |>
  mutate(
    is_current = ifelse(loan_status == "Current", 1, 0))

df_test <- df_clean |>
  filter(home_ownership %in% c("OWN", "RENT", "MORTGAGE")) |>
  mutate(
    home_group = case_when(
      home_ownership %in% c("OWN", "MORTGAGE") ~ "Owner/Mortgage",
      home_ownership == "RENT" ~ "Rent"
    ),
    emp_length = gsub("[^0-9]", "", emp_length),
    emp_length = as.numeric(emp_length))
```

EDA

```r
# Income
# Create Income Brackets
income_summary <- df_test |>
  mutate(
    income_bracket = case_when(
      annual_inc < 56000 ~ "lower",
      annual_inc < 165000 ~ "middle",
      TRUE ~ "upper"
  )) |> group_by(income_bracket) |>
  summarise(percent_current = mean(is_current)*100)
```

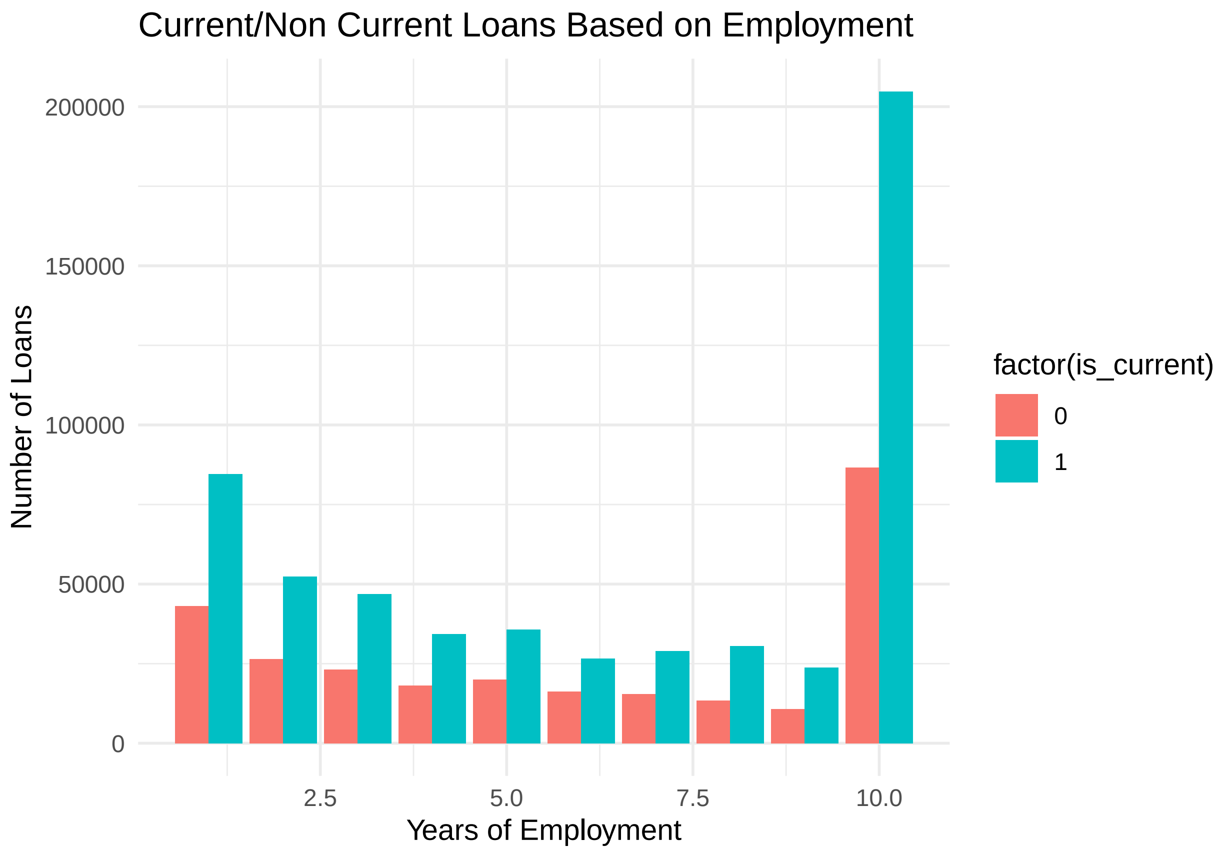
```
ggplot(income_summary, aes(x = income_bracket, y = percent_current)) +
  geom_segment(aes(x = income_bracket, xend = income_bracket, y = 60, yend = percent_current, color = in
  geom_point(size = 4) +
  geom_text(aes(label = paste0(round(percent_current, 1), "%")), vjust = -1) +
  coord_cartesian(ylim = c(60,75)) +
  labs(
    title = "Percent of Current Loans Based on Income Bracket",
    x = "Income Bracket",
    y = "Percent Current"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
# Employment
emp_summary <- df_test |>
  group_by(emp_length, is_current) |>
  summarise(n = n(), .groups = "drop")

ggplot(emp_summary, aes(x = emp_length, y = n, fill = factor(is_current))) +
  geom_col(position = "dodge") +
  labs(
    title = "Current/Non Current Loans Based on Employment",
    x = "Years of Employment",
    y = "Number of Loans"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_col()`).
```



Statistical Testing

```r
# How often each group stays current
summary_table <- df_test |>
  group_by(home_group) |>
  summarise(
    total = n(),
    percent_current = round(mean(is_current)*100, 1)
  )

summary_table
```

```
## # A tibble: 2 x 3
##   home_group      total percent_current
##   <chr>           <int>           <dbl>
## 1 Owner/Mortgage 531027            68.9
## 2 Rent           356117            66.3
```

```r
# Chi Squared Test
test_table <- table(df_test$home_group, df_test$is_current)

chisq.test(test_table) # very small p-value implies significant difference
```

```
##
```

```
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  test_table
## X-squared = 673.16, df = 1, p-value < 2.2e-16
```

```r
# Logistic Regression
df_model <- df_test |>
  mutate(
    is_current = as.numeric(is_current),
    home_group = factor(home_group)
    )  |>
  filter(
    !is.na(is_current),
    !is.na(home_group),
    !is.na(annual_inc),
    !is.na(emp_length))

# model only for home owenership
model_simple <- glm(is_current ~ home_group,
data = df_model,
family = "binomial")


# model with employment and income
model_full <- glm(is_current ~ home_group + emp_length + annual_inc,
             data = df_model,
             family = 'binomial')

summary(model_simple)
```

```
##
## Call:
## glm(formula = is_current ~ home_group, family = "binomial", data = df_model)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)     0.779253   0.003041   256.3   <2e-16 ***
## home_groupRent -0.116713   0.004725   -24.7   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1062144  on 842322  degrees of freedom
## Residual deviance: 1061535  on 842321  degrees of freedom
## AIC: 1061539
##
## Number of Fisher Scoring iterations: 4
```

```r
summary(model_full)
```

```
##
## Call:
## glm(formula = is_current ~ home_group + emp_length + annual_inc,
##     family = "binomial", data = df_model)
```

```
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)      5.520e-01  6.684e-03   82.59   <2e-16 ***
## home_groupRent -6.580e-02  4.912e-03  -13.39   <2e-16 ***
## emp_length      1.873e-02  6.729e-04   27.83   <2e-16 ***
## annual_inc      1.236e-06  5.055e-08   24.45   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1062144  on 842322  degrees of freedom
## Residual deviance: 1060033  on 842319  degrees of freedom
## AIC: 1060041
##
## Number of Fisher Scoring iterations: 4
```

```r
# Compare models
anova(model_simple, model_full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: is_current ~ home_group
## Model 2: is_current ~ home_group + emp_length + annual_inc
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1    842321    1061535
## 2    842319    1060033  2   1502.1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visual For Likelihood Ratio Test

```r
lrt_df <- data.frame(
  model = c("Home Only", "Full Model"),
  deviance = c(1061535, 1060033)
)

ggplot(lrt_df, aes(x = model, y = deviance)) +
  geom_col(fill = "steelblue") +

  # Looked up how to add a line to visually show the decrease in deviance
  geom_segment(aes(x =1.5, xend = 1.5,
                   y = 1060033, yend = 1061535),
               color = "darkorange", size = 1.2) +
  annotate("text", x =1.6, y = (1061535 + 1060033)/2,
           label = "1502",
           color = "Brown",
           hjust = 2) +

  coord_cartesian(ylim = c(1055000, 1063000)) +
  geom_text(aes(label = round(deviance, 0)), vjust = -0.5) +
  labs(
    title = "Model Fit Comparison",
```

```r
    subtitle = "Lower deviance indicates better fit",
    x = "Model",
    y = "Residual Deviance"
  ) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning in geom_segment(aes(x = 1.5, xend = 1.5, y = 1060033, yend = 1061535), : All aesthetics have
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```